

Jihai Zhao

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EDUCATION

Northwestern University

M.S. in Robotics

Sept. 2023 – Dec. 2024

The Ohio State University

B.S. with Honors in Mechanical Engineering and Minor in Computer Science

Sep. 2019 - May. 2023

SKILLS

Languages:	C++, Python, C, MATLAB/Simulink, R
AI/ML:	PyTorch, JAX, Diffusion Policy, Imitation Learning, VLA Models, Computer Vision (OpenCV, Stereo Vision), Transformers
Robotics:	Robot Operating System(ROS/ROS 2), MoveIt, Dora, IsaacSim, PyBullet, Gazebo, Rviz, Urdf/Xacro, SLAM (EKF), Path Planning (RRT, A*), Kalman Filter
Software:	Linux, Docker, CMake, Git, Unit Testing
Hardware:	SolidWorks, OnShape, Rapid Prototyping, 3D printing

WORK EXPERIENCE

Stealth Startup - Robotics Software Engineer

Jan. 2025 – Dec. 2025

- Fine-tuned state-of-the-art VLA models (pi0, pi0.5) for dual xarm7 setup, achieving state-of-the-art performance on long-horizon manipulation tasks.
- Engineered a synthetic data collection pipeline utilizing mask-tracking (SAM2 and CUTIE) to augment datasets.
- Enhanced diffusion policy by incorporating segmentation targets' masks, achieving higher task success rates in object picking and placement.
- Architected a real-time perception stack integrating FoundationPose and Grounded-SAM to achieve sub-centimeter 6D grasping precision.

Realman-Robotics - Robotics Software Engineer Intern

Jun. 2024 – Sept. 2024

- Implemented a real-time S-curve velocity planner in C++, minimizing motion jerk and improving smoothness.
- Designed a stereo vision self-calibration routine involving spherical trajectory optimization and least-squares kinematic parameter adjustment, reducing absolute positioning error by 40% in the target workspace.

PROJECT

Generative AI for Robot Manipulation

Sept. - Dec. 2024

- Implemented a VLM-driven high-level planner that parses natural language instructions (image + text) into executable robot primitives, bridging the gap between semantic understanding and low-level control.
- Fused SAM segmentation masks with DINOv2 patch embeddings to cluster scene objects via k-means.
- Utilized 3D point cloud data to compute grasp poses, ensuring robust interaction with objects in diverse environments.

Simultaneous Localization and Mapping (SLAM) from Scratch

Jan. – Mar. 2024

- Engineered a custom SLAM system from scratch in C++ (ROS2) for a differential drive robot, implementing Extended Kalman Filter (EKF) state estimation and landmark data association without reliance on external mapping libraries.
- Developed a C++ package for differential drive kinematics, collision detection, 2D transformations, and odometry.
- Implemented a landmark detection algorithm through supervised learning.
- Used C++ to implement a custom simulation environment for the robot, including lidar, obstacles, and collisions.

Autonomous Manipulation with Franka Emika

Sept. – Dec. 2023

- Worked in a group of five to develop ROS2 packages to control the Franka robot arm to brew a cup of coffee.
- Used Computer Vision (OpenCV and AprilTags) to find the location of each object and wrote a custom wrapper package for MoveIt in Python to control the robot.

KUKA YouBot Manipulation

Sept. – Dec. 2023

- Planned a trajectory for the end-effector of a mobile base with four mecanum wheels and a 5R robot arm by using feed-forward control and a PI controller.
- Performed feedback control to drive the YouBot to pick up a block at a specified location, carry it to a desired location, and put it down. It was simulated in CoppeliaSim.

Vision Transformer for Surface Defect Inspection

Jun. - Oct. 2022

- Applied Vision Transformer (VT) model to surface defect detection. The VT model combines image linear projection, position embedding as the input, and multilayer perceptron (MLP) as the classifier.
- Achieved 93.3% accuracy and 0.9302 F1 score on the dataset of aluminum surface flaws, demonstrating it performs better than other CNN baselines (including VGG19, DenseNet, and ResNet).